# Python Lab exercise- 5

1. Write a program to calculate simple interest using a function interest() that can receive principal amount, time and rate and returns calculated simple interest. Do specify default values for rate and time as 10% and 2 years respectively.
2. Write a program that receives 2 numbers in a function and returns the results of all arithmetic operations (+, -,*,/ ,%) on these numbers.
3. Write a function that takes a positive integer and returns the one's position digit of the year
4. Write a function that takes amount-in-dollars and dollar-to-rupee conversion price;it returns the amount converted to rupees. Create the function in both void and non-void forms.
5. Write a program to have following functions
   a. a function that takes a number as argument and calculates cube for it. If there is no value passed to the function in function call, the function should calculate cube of 2.
   b. a function that takes 2 char arguments and returns True if both the arguments are equal otherwise False
6. Write a function that receives 2 string arguments and checks whether they are same length strings ( returns True otherwise False)
7. Write a Python program using a lambda function to find the square of a given number.
8. Write a lambda function that takes a number and returns "Even" if it is even and "Odd" otherwise.
9. Write a lambda function to find the greater of two numbers.
10. Write a lambda function to convert temperature from Celsius to Fahrenheit.
11. Write a lambda function to calculate simple interest given the principal, rate, and time.
12. Given a list of integers, use filter() and a lambda function to create a list containing only the even numbers.
13. Given a list of numbers, use a lambda function with `filter()` to extract numbers greater than 50.
14. Use map() and a lambda function to generate the squares of all numbers in a list.
15. Given a list of strings, use `map()` and lambda to convert every string to uppercase.
16. Given a list containing positive and negative numbers, use `filter()` and lambda to extract only the positive numbers.
17. Write a Python program using regular expressions to find all digits present in a given string.
18. Write a program to find all words beginning with the letter A or a.
19. Write a program to check whether the word "Python" occurs in a given sentence.

20. Given a string containing numbers and other characters, extract all the integer numbers using regular expressions.
21. Write a Python program to create a class BankAccount with the following requirements:
    The class should have the following attributes:
    account_number
    holder_name
    balance
    a. Create a constructor to initialize these attributes.
    b. Define the following methods:
        (i)deposit(amount) – Add money to the account.
        (ii)withdraw(amount) – Withdraw money only if sufficient balance is available.
        (iii)display_balance() – Display the current balance.
        (iv)display_details() – Display account number, holder name, and balance.
    c. Create at least three objects of the BankAccount class.
    d. Perform different deposit and withdrawal operations on each account.
    e. Display the details of all accounts at the end.
22. Write a Python program to create a class Student. The class should contain:
    roll_no
    name
    marks in 3 subjects
    Define methods to:
        a. Calculate the total marks.
        b. Calculate the average marks.
        c. Determine the grade based on the average:
            - 90 and above → A+
            - 80–89 → A
            - 70–79 → B
            - 60–69 → C
            - 50–59 → D
            - Below 50 → F
        d. Display the complete student details along with total, average and grade.

Create objects for 5 students and display their results.

23. Write a Python program to create a class Book with the following attributes:
    book_id
    title
    author

price
available — indicates whether the book is available for borrowing.

Create methods to:

a. Display the details of a book.
b. borrow_book() – Allow a student to borrow the book only if it is available.
c. return_book() – Mark the book as available when it is returned.
d. Display the current availability status.

Create at least 5 book objects and perform different borrowing and returning operations.

24. Create a class Employee with the following data members:
Employee ID
Employee name
Department
Basic salary
Number of years of service

Define methods to:

a. Calculate HRA as 20% of basic salary.
b. Calculate DA as 10% of basic salary.
c. Calculate a bonus based on years of service:
    5 years or more → ₹10,000
    3–4 years → ₹5,000
    Less than 3 years → ₹2,000
d. Calculate gross salary.
e. Calculate tax as 10% of gross salary if gross salary exceeds ₹50,000.
f. Calculate net salary.
g. Display the employee and salary details.

Create objects for 5 employees and determine the employee receiving the highest net salary.

25. Create a class Consumer containing:
Consumer number
Consumer name
Previous meter reading
Current meter reading

Define methods to:

a. Calculate units consumed.
b. Calculate electricity charges using:

| Units | Rate |
|---|---|
| First 100 | ₹2/unit |
| 101–200 | ₹3/unit |
| 201–500 | ₹5/unit |
| Above 500 | ₹7/unit |

.

c. Calculate 5% surcharge if the bill exceeds ₹2,000.
d. Display the complete bill.

Create objects for several consumers and call the methods